Apartment & Roommate Finder

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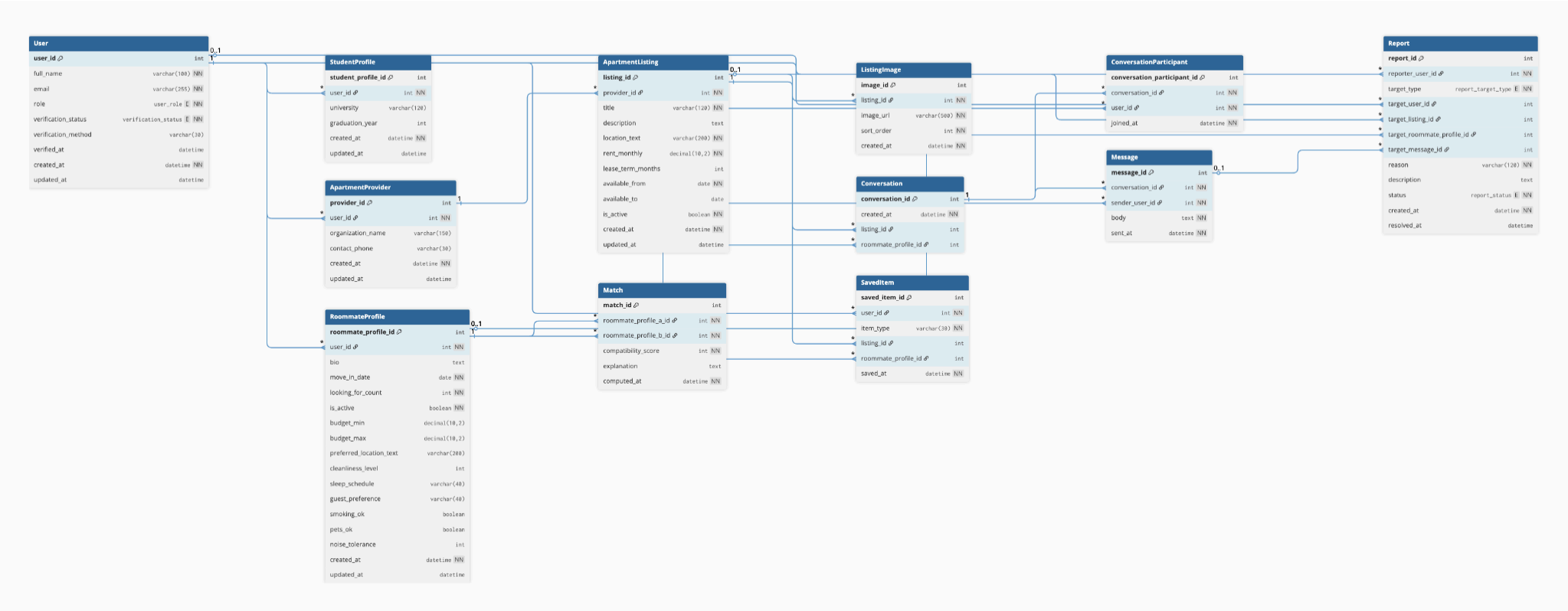
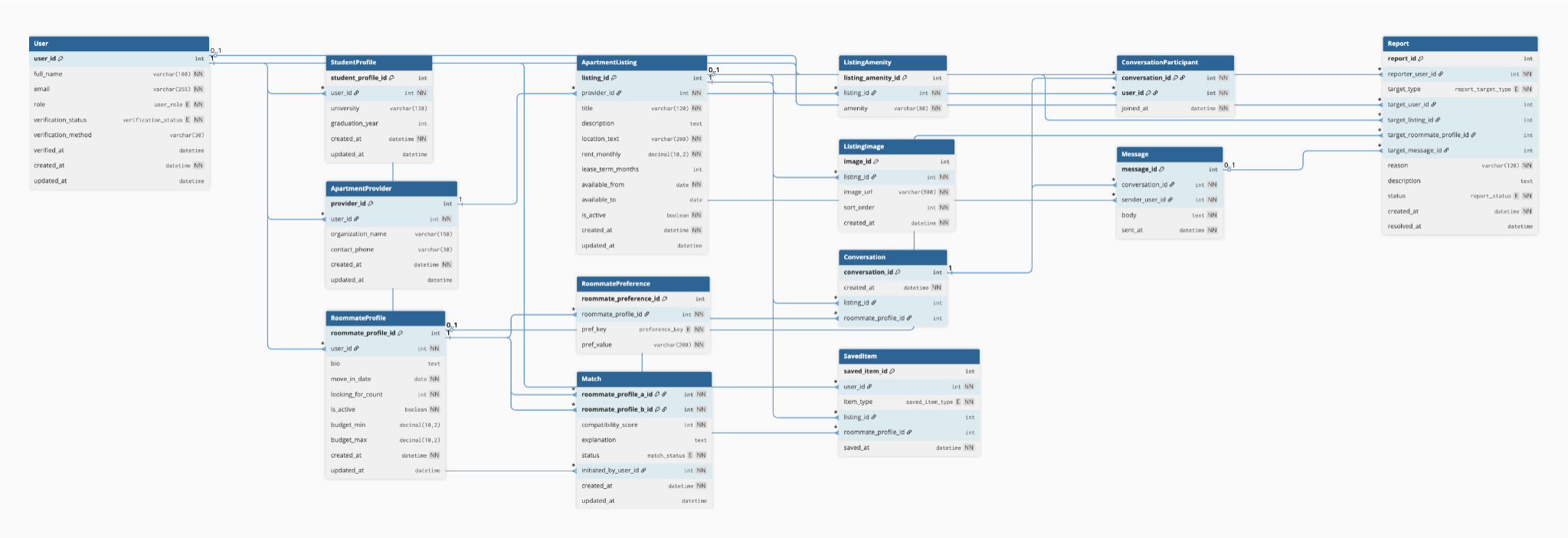
Previous Phase 2 Entity Relationship Diagram



Updated Entity Relationship Diagram (also Relational Schema Diagram)



[Interactive ERD (higher quality)]

Changes Made

For Phase 3, we converted the many-to-many relationships in our conceptual ER diagram into associative relations in the logical design. The compatibility relationship between two RoommateProfile entities is implemented using the Match relation with a composite primary key (roommate_profile_a_id, roommate_profile_b_id), which prevents duplicate match pairs and ensures each match represents a relationship between two distinct roommate profiles. Likewise, the many-to-many relationship between User and Conversation is implemented using the ConversationParticipant relation with a composite primary key (conversation_id, user_id), preventing duplicate membership entries.

We explicitly defined primary keys and foreign keys for every relation in the logical schema. Each table clearly identifies its primary key, and foreign keys reference their parent relations to enforce referential integrity. One-to-one relationships, such as between User and StudentProfile, User and ApartmentProvider, and User and RoommateProfile, are enforced using unique constraints on the foreign key attributes so the logical schema matches the intended cardinalities.

To incorporate the TA feedback and improve normalization and matching queries, we introduced separate Preference-type entities instead of embedding compatibility fields inside profiles. Specifically, we added a RoommatePreference relation to store lifestyle and location preferences in a structured way, enabling efficient filtering and compatibility scoring without relying on embedded fields. We also introduced a ListingAmenity relation so amenities are modeled as a proper multi-valued relationship rather than being stored inside a listing record.

Finally, we enhanced the Match relation to better represent evolving roommate decisions rather than a static compatibility link. In addition to compatibility_score and explanation, Match now includes a status attribute (pending, accepted, declined), timestamps, and the initiating user, allowing the system to track roommate interactions over time while maintaining data integrity. These refinements ensure the logical database design is cleaner, more queryable, and better aligned with real workflows and relational design principles.

Roles of Project Members

Daniel Carrillo

Role:

- Contribute to database schema design and refinement, including entity relationships and constraints
- Assist in defining system requirements and translating them into technical specifications
- Design and document value-added system features with a focus on scalability and data integrity
- Support backend development by validating queries, constraints, and data workflows
- Contribute to testing and verification to ensure correctness and consistency across system components
- **Worked on System Users and Privileges - Phase 2**
- **Contributed to ER diagrams' new implementation - Phase 3**

Hursh Soni

Role:

- Assist with DB design and implementation (data models, relationships, etc.)
- Work on frontend design & development
- Assist with backend development
- Work on integration testing
- Project documentation through the semester
- **Worked on ERD and updates - Phase 3**
- **Documentation - Phase 3**

Anthony Osei

Role:

- Work on backend development
- Work on project phase documentations
- Assist with database design and implementation
- **Contributed to creating ER diagrams, changes made - Phase 3**

Vivan Verma

Role:

- Work with Anthony to ensure a well-developed backend
- Work with Hursh to also develop a frontend
- Testing
- **Worked on problem definition entities and their description and relationships - Phase 2**
- **Worked with AI to clean up tasks, help with documentation, and ERD updates - Phase 3**
- **Worked with Hursh on Documentation - Phase 3**

Generative AI

Role:

- Help with debugging/identifying issues
- Assist with system architecture and design
- Help understand the project scope and provide ideas
- **Clean up tasks, help with documentation, and ERD updates - Phase 3**
- **Feedback and final edits - Phase 3**
- **Arranging ER - Phase 3**